

# Student Guide — senior / module-15- project-implementation

IKS Curriculum

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## Student Workbook

### Student Workbook — Project Implementation — Capstone (Senior)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

### Vocabulary tracker

Fill in as we go:

- *anvayana* — \_\_\_\_\_
- *pañcagavya* — \_\_\_\_\_
- *siddhānta* — \_\_\_\_\_

### Day-by-day notes

#### Day 1 — Capstone Overview and Project Menu

**What I learned:**

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**One question I have:**

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#### Day 2 — Scope and Plan

**What I learned:**

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**One question I have:**

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Day 3 — Research — Day 1

**What I learned:**

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**One question I have:**

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Day 4 — Research — Day 2

**What I learned:**

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**One question I have:**

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Day 5 — Mid-Project Critique

**What I learned:**

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**One question I have:**

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Day 6 — Build / Make — Day 1

**What I learned:**

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**One question I have:**

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## Day 7 — Build / Make — Day 2

**What I learned:**

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**One question I have:**

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## Day 8 — Document

**What I learned:**

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**One question I have:**

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## Day 9 — Rehearse Presentation

**What I learned:**

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**One question I have:**

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## Day 10 — Project Share Day

**What I learned:**

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**One question I have:**

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## Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

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# Day 1 — Capstone Overview and Project Menu

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

## Learning objective

By the end of this lesson, students will: review the 10-option project menu; pick a project.

## Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

## Vocabulary introduced today

- *anvayana* (IAST: anvayana; lit. “integration / following-through; capstone synthesis”)

## Primary source (today’s reading)

From *All Modules 1–14 of this curriculum* — see `sources.md`.

## Lesson flow

### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

### Core (20 min)

1. **Capstone overview.** This module is different. No new content delivery. Instead: 10 days of structured project work, applying what you learned from Modules 1–14.
2. **Walk through the 10-option project menu.** Read each, with a 1-minute description per option:
  - Pañcagavya kitchen science → SUBSTITUTED with fermented-foods study (safety)
  - Mandala / Pṛthvī geometry
  - Tithi/festival calendar
  - Herb identification card-deck
  - Magic-square art
  - Yoga sequence design
  - Sustainability audit
  - Doshic self-study

- Multiplication-method comparison
- Open-design (your own)
- 3. **The discipline.** Each project requires: scope, research, design, build, document, present. Daily journal. Source citations.
- 4. **The safety floor.** Pañcagavya — NEVER cow-urine ingestion in school. Use comparative fermented-foods study (curd, paneer, batter — biochemically interesting and safe).
- 5. **Today’s task.** Read all 10 options in detail (printout provided). DON’T pick yet — sleep on it. Tomorrow you’ll choose and scope.

### Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Capstone Overview and Project Menu.

### Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

### Homework

(See `homework.md` for today’s task.)

### Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

### Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

## Day 2 — Scope and Plan

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

### Learning objective

By the end of this lesson, students will: write a 1-page project plan: problem, sources, deliverable, deadline.

### Materials

- Whiteboard / chart paper
- Notebook, pen per student

- (See lesson-plan.md for any day-specific materials)

## Vocabulary introduced today

- *pañcagavya* (IAST: pañcagavya; lit. “five-cow-product preparation (substituted with fermented-foods study for school safety)”)

## Lesson flow

### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

### Core (20 min)

1. **Each student commits to ONE project.** Show of hands per option to see distribution.
2. **The one-page plan template.**
  - Title (~10 words)
  - Problem (~50 words): what question/need does this address?
  - Sources (~5 listed): the modules + texts you’ll draw from
  - Deliverable (~50 words): what will exist at the end?
  - Timeline (Day 3-10 broken down by phase)
  - Success criteria (3 bullet points: what makes this excellent?)
3. **Work-time.** 30 minutes of silent plan-writing.
4. **Pair-share.** Pairs swap plans, give one strength + one concern (sticky notes).
5. **Submit.** Plans go to the teacher tonight for sign-off. Tomorrow you begin research.

### Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Scope and Plan.

### Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

## Homework

(See [homework.md](#) for today’s task.)

## Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

## Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

## Day 3 — Research — Day 1

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

### Learning objective

By the end of this lesson, students will: gather sources; cross-reference earlier modules.

### Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

### Vocabulary introduced today

- *siddhānta* (IAST: *siddhānta*; lit. “concluded principle or doctrine”)

### Lesson flow

#### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

#### Core (20 min)

1. **Research phase begins.** Each student returns to the relevant Modules for content:
  - Doṣa project → Module 4
  - Herbs project → Module 10
  - Astronomy project → Module 6 or 12
  - Math project → Module 5, 7, 9, 11, or 14
  - Ecology project → Module 13
2. **The library / research routine.** Open the relevant pack’s *sources.md*. Open at least 2 cited sources (online or print). Take 2-page notes per source. The notes will support the project’s references.
3. **Critical eye.** Don’t just trust the FIRST source. For each claim, ask: “*Who’s the author? When was this written? What’s their context?*”
4. **Work-time.** 35 minutes of silent research + note-taking.

5. **Sharing (5 min).** Three students briefly describe what they learned and what surprised them.

### Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Research — Day 1.

### Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

### Homework

(See [homework.md](#) for today’s task.)

### Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

### Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

## Day 4 — Research — Day 2

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

### Learning objective

By the end of this lesson, students will: continue research; sketch initial design.

### Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

### Vocabulary introduced today

- *anvayana* (IAST: *anvayana*; lit. “integration / following-through; capstone synthesis”)



## Lesson flow

### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

### Core (20 min)

1. **Continued research.** Today’s focus: sketches + design.
2. **The design-thinking question.** “*What will this LOOK like when it’s done?*” Draw it. Even a rough sketch helps you find missing pieces.
3. **Source check.** Revisit yesterday’s notes. Are there gaps? Look up one more source if needed.
4. **Work-time.** 30 minutes of sketching + planning the build.
5. **Pair-feedback.** Pairs discuss design. “*What’s the strongest part of your design? What’s still unclear?*”

### Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Research — Day 2.

### Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

## Homework

(See [homework.md](#) for today’s task.)

## Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

## Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

## Day 5 — Mid-Project Critique

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

## Learning objective

By the end of this lesson, students will: pairs swap plans; offer one strength + one concern.

## Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

## Vocabulary introduced today

- *pañcagavya* (IAST: pañcagavya; lit. “five-cow-product preparation (substituted with fermented-foods study for school safety)”)

## Lesson flow

### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

### Core (20 min)

1. **Critique circle structure.** Pairs swap project plans + sketches. 5 minutes silent reading. Then each gives:
  - One strength (sticky note).
  - One concern (sticky note).
  - One question (verbal).
2. **Pairs rotate twice.** Each student gets feedback from 3 different partners.
3. **Synthesise.** Each student writes a 100-word “what I’m changing” note based on critique. May include:
  - Narrowing scope
  - Adding a source
  - Clarifying deliverable
  - Revising timeline
4. **Honest about limits.** “*What’s still hard? What might fail? What’s your Plan B?*” Real projects always have unknowns. Acknowledge them.
5. **Submit revised plan.** Teacher reviews tonight. Day 6 = build phase begins.

### Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Mid-Project Critique.

### Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

### Homework

(See `homework.md` for today’s task.)

### Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

### Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

## Day 6 — Build / Make — Day 1

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

### Learning objective

By the end of this lesson, students will: start building the artefact (model, poster, deck, garden plot).

### Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

### Vocabulary introduced today

- *siddhānta* (IAST: *siddhānta*; lit. “concluded principle or doctrine”)

### Lesson flow

#### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

#### Core (20 min)

1. **Build phase begins.** Materials should be ready (announced last week). If not, today is also a “gather materials” day.

## 2. Project-specific guidance per option.

- Magic-square art → grid + colours + calligraphy.
- Herb deck → photographs / drawings + card layout + lamination.
- Yoga sequence → flow design + safety check + pose drawings.
- Sustainability audit → measurement instruments + spreadsheet.
- And so on.

3. **Daily journal.** Today's entry: "*What did I build today? What worked? What didn't?*"

4. **Work-time.** 35 minutes of focused making.

5. **End-of-day share (5 min).** Three students share progress.

## Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Build / Make — Day 1.

## Wrap-up (5 min)

- One-sentence exit ticket: "*What's one thing you learned today?*"
- Preview tomorrow.

## Homework

(See [homework.md](#) for today's task.)

## Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

## Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

# Day 7 — Build / Make — Day 2

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

## Learning objective

By the end of this lesson, students will: continue building; daily journal entry.

## Materials

- Whiteboard / chart paper

- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

## Vocabulary introduced today

- *anvayana* (IAST: anvayana; lit. “integration / following-through; capstone synthesis”)

## Lesson flow

### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

### Core (20 min)

1. **Continue building.** Focus on completing the major deliverable.
2. **Journal entry today:** “*What progress? What’s slowed me down? What do I still need to do tomorrow?*”
3. **Mid-build review.** Teacher checks each student’s progress. Flags students who may need help in time-management or scope.
4. **Honest scope adjustment.** If you realise you can’t finish in time, narrow the scope NOW (not on Day 9). Better to do less, well, than more, poorly.
5. **Independent work-time.** 40 minutes. Quiet, focused.

### Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Build / Make — Day 2.

### Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

## Homework

(See [homework.md](#) for today’s task.)

## Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

## Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.

- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

## Day 8 — Document

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

### Learning objective

By the end of this lesson, students will: write up process, sources, reflections (~500 words for senior).

### Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

### Vocabulary introduced today

- *pañcagavya* (IAST: *pañcagavya*; lit. “five-cow-product preparation (substituted with fermented-foods study for school safety)”)

### Lesson flow

#### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

#### Core (20 min)

1. **Documentation phase.** Today: write up your project.
2. **The structure (Senior — ~500 words).**
  - **Problem & motivation** (~100 words): what did you set out to do?
  - **Sources used** (~100 words): which Modules, which texts, which observations?
  - **Process** (~150 words): what did you actually do? Day by day?
  - **Findings / outcomes** (~100 words): what did you produce / observe?
  - **Reflection** (~50 words): what surprised you? What would you do differently?
3. **Honest reflection rule.** Include at least ONE thing that didn’t work. Real research has failures. Including yours is intellectually honest — and pedagogically valuable.
4. **Citations.** Use the format: *Author, Title, Year*. See pilot modules’ `sources.md` for examples.
5. **Drafting time.** 35 minutes silent writing. Teacher available for questions.

## Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Document.

## Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

## Homework

(See [homework.md](#) for today’s task.)

## Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

## Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

# Day 9 — Rehearse Presentation

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

## Learning objective

By the end of this lesson, students will: pair feedback on final presentation; refine.

## Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

## Vocabulary introduced today

- *siddhānta* (IAST: *siddhānta*; lit. “concluded principle or doctrine”)

## Lesson flow

### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

## Core (20 min)

1. **Pair-rehearsal.** Pairs swap presentations. Time the 5-minute talk. Friend gives:
  - One strength (clear, well-organised, engaging).
  - One specific suggestion (slow down, add an example, more eye contact).
2. **Refine.** Each student adjusts their presentation based on feedback. 10 minutes solo refinement.
3. **Group-rehearsal (5 students share with class).** Selected by teacher (varied projects). Class gives 2 minutes of feedback each.
4. **Logistics check.** Materials ready for tomorrow? Backup plan if something breaks?
5. **Self-assessment.** Each student rates themselves on the rubric (assessments/rubric.md). They submit this PRE-presentation; the teacher will compare with their own scoring after the share.

## Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Rehearse Presentation.

## Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

## Homework

(See [homework.md](#) for today’s task.)

## Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

## Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

# Day 10 — Project Share Day

**Module:** Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min



## Learning objective

By the end of this lesson, students will: present project to peers + parents; complete self-assessment.

## Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

## Vocabulary introduced today

- *anvayana* (IAST: anvayana; lit. “integration / following-through; capstone synthesis”)

## Lesson flow

### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

### Core (20 min)

1. **Project share day.** Each student presents their project (5 minutes including Q&A). Audience: peers, optionally parents, optionally other teachers.
2. **Format.** Brief: what + why + how + what-I-learned. Show the artefact. Take 1-2 questions.
3. **Audience role.** Each peer takes one note per project (1 sticky note, one observation or compliment). At the end, sticky notes go to each presenter.
4. **Self + teacher assessment.** After the share, each student completes their final self-assessment on the rubric. The teacher fills the corresponding teacher-assessment. Within a week, the two are compared in a 1-1 conversation.
5. **Module close (5 min).** Whole-class reflection: “*Looking back across all 15 modules — Modules 1-14 of curriculum, Module 15 of project — what’s ONE thing that will stay with you?*” Each student says one sentence.
6. **Curriculum close.** Brief teacher comment + final round of applause. The IKS curriculum is complete.

### Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Project Share Day.

## Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

## Homework

(See `homework.md` for today’s task.)

## Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

## Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

## Writing Prompts

### Writing Prompts — Project Implementation — Capstone (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Project Implementation — Capstone to a friend who hasn’t taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Project Implementation — Capstone with a similar concept from another module or from your science textbook. What’s similar? Different?
4. Identify ONE claim in this module that you’d want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Project Implementation — Capstone? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji’s mathematics genuinely Vedic? are *doṣas* a valid framework today?). Take a position with two pieces of evidence.

## How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

## Homework

### Homework — Project Implementation — Capstone (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Capstone Overview and Project Menu** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Scope and Plan** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Research — Day 1** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Research — Day 2** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Mid-Project Critique** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Build / Make — Day 1** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Build / Make — Day 2** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Document** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Rehearse Presentation** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Project Share Day** · Write 2 sentences about today's concept. (~10 min)

## Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.